

# Xiaoxiao YU

Agent Engineer | Full-Stack Engineer

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Full-stack agent engineer who ships production AI products end to end, from React/TypeScript frontends and Python/Node.js backends to the agent runtime underneath. At Viciking (full-time since July 2026), built the agent products that **doubled video-production efficiency**. Before: a browser-agent startup in Ant Group's MaShang AI Accelerator, zkWasm apps with thousands of users, and React Native apps.

## Experience

### Viciking (Beijing Weixi Network Technology)

Jul 2026 – Present

*Agent Engineer, full-time* | *sole engineer, frontend to agent runtime, on four internal AI production products*

- Versus manual editing, **production efficiency doubled**; the tools serve dozens of internal users across multiple business lines.
- WeiChuang Drama Studio** (React 19, FastAPI, Temporal, PostgreSQL): a 1-minute episode goes **from script to finished video in 15–30 min**, with episodes in parallel; **~90% need no manual re-editing**, the rest only light edits. If a model call fails mid-batch, only the failed shots rerun, so finished shots are never regenerated or paid for twice. Bad generations fall back to another model or get their voice repaired, and every model call is logged so a slow or stuck job can be traced to the exact call. ~5,000 tests.
- Weixi Studio** (TypeScript, React, Remotion, PostgreSQL, MCP): turned OpenChatCut, a local-first single-user agentic video editor, into a **multi-tenant SaaS used across business lines**. Agent tasks survive worker crashes and resume where they stopped; a paid or irreversible tool call is never retried blindly, and if its outcome is unclear a person decides. Over MCP, outside agents and people edit one timeline without overwriting each other.
- Screenplay Agent Studio** (FastAPI, Pi agent SDK, OpenTelemetry): built an eval comparing one lead agent with a lead plus specialists; the specialists cost **3.7× the tokens for no quality gain**, so each of the 6 writing stages runs a single lead. Drafts live in files rather than chat history, so long sessions can be compressed without losing work; writers approve every edit; alerts fire when agent runs start failing or a decision waits too long.
- Creative Engine** (TanStack Start, pgvector): a writing agent that drafts from analyzed reference videos and keeps what it observed apart from what it inferred; ran 14 controlled experiments with an LLM judge and removed every prompt technique that did not measurably help.
- Clip pipeline (team project)**: ported Weixi Studio's AI pre-roll, scene recreation, watermark removal and stitching into the team's v2 pipeline, which turns dramas into ad clips at scale, and built the unattended uploader that ships them to the ad platform, work previously done by hand, with no duplicate submissions. Separately built a GPU redaction pipeline (fine-tuned RF-DETR) that masks sensitive objects at **2× real time**, ~80% accurate.

### Delphinus Lab

Nov 2024 – Jul 2026

*Developer Relations Engineer* → *Ecosystem Director (Jun 2025)* | *Remote, concurrent with MPhil*

- Rebuilt the zkWasm browser SDK from scratch so apps use one wallet connection across both chain layers; wrote the CLI that scaffolds new apps; maintained the TypeScript services that settle deposits and withdrawals.
- Solo-built staking, launchpad, prediction-market and social-binder apps, plus the zkWasm Hub site and developer docs; ran DevOps for 8+ ecosystem apps, which reached **thousands of users** and **multi-million-USD TVL**.
- Led the team that built a hosted Kubernetes platform where developers deploy straight from GitHub and watch build status and logs live; it hosts **10+ projects**.

### Earlier roles (2021–2024, alongside undergraduate studies)

- Rosen** — Founding Full-Stack Engineer (2022–23): React / React Native apps for iOS, Android and Web; backed by PayPal and Circle. **2PM.Network** — Co-Founder & CEO (2024). **HashPunk** — Product Manager (2021–22), raised US\$1M. **1CIRCLE** — Full-Stack Engineer (2021–22). **Conflux** — Intern (2021).

## Selected Projects

### Resona — Browser Agents for Product Demos

May – Jun 2026

*Own startup, sole engineer* | *Ant Group MaShang AI Accelerator; trial customers* | *Stagehand, Playwright, AgentBay, BullMQ*

- Give it a product URL and it returns a demo video: browser agents map the product, explore each section in parallel in cloud sandboxes, track what they have covered and go back for anything missed. Any action that changes data needs a check first, and payments are blocked outright.
- When an agent hits a login page, the user signs in once in the same browser and the agent carries on from there. Explore, record and render run as separate job queues; ~500 tests.
- Render agents (GSAP, Remotion) cut the recordings into the finished demo; each step picks its own model from OpenAI, Anthropic or Gemini.

- Designed 17 role agents and 25 skills that take a novel through ingest, analysis, episode planning, writing, review and storyboard / video prompts. A review director gates every episode, and failures loop back for rewrite and re-review.
- Progress is saved to files, so a run can stop and resume at any episode without the context window filling up; hybrid ChromaDB + TF-IDF search pulls similar reference scripts to ground the writing.

ShotLoom — AI Film Shot-Breakdown Workspace

Open source, sole author | FastAPI, Redis/RQ, React, TypeScript, Qwen vision + ASR

- Built a shot-by-shot film breakdown: scene detection and clip cutting, then Qwen vision analysis of shot size, camera movement, lighting and narrative function, aligned with ASR transcripts and a whole-film continuity report.
- Long films are split into overlapping chunks so no scene is lost at a boundary; progress streams live and survives a page refresh; Excel/PDF export; 107 end-to-end tests.

RingChat — Agent-Native Messaging

Jan – Mar 2026

Team lead, client engineer | Kotlin / Jetpack Compose, Web, iOS | OpenClaw (self-hosted agent gateway)

- Led a **6-person team** building a cross-platform IM in which every user gets a personal AI agent, and worked mainly on the clients (now continued by another team as Gold House). On Android, built the chat UI, optimistic media upload, contact and group search, unread badges and multi-device sync.

Skills

|                   |                                                                                                                                                                  |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Agents            | Context engineering, tool and MCP design, multi-agent orchestration, browser agents (Stagehand, Playwright, AgentBay), human-in-the-loop, evals and LLM-as-judge |
| Frontend & mobile | TypeScript, React 19, TanStack (Start / Query), Next.js, Vite, Remotion; React Native, Kotlin (Compose)                                                          |
| Backend           | Python (FastAPI), Node.js; PostgreSQL / pgvector, Redis, Temporal, BullMQ, SSE, WebSocket                                                                        |
| Infra & models    | Docker, Kubernetes, CI/CD, OpenTelemetry; OpenAI, Anthropic, Gemini, Qwen, Doubao/Seedance via Vercel AI SDK and gateways                                        |
| Other             | Solidity, Rust, C++ (ESP32), PyTorch, scikit-learn                                                                                                               |

Education

The Hong Kong University of Science and Technology (Guangzhou)

Sep 2024 – Oct 2026

Red Bird MPhil in Data Science and Analytics | graduated Oct 2026 | thesis: local-first retrieval over 1.36M Mastodon posts, 79.3% answer accuracy at under 100 ms per query

The University of Edinburgh

Sep 2020 – Jun 2024

MA in Cognitive Science, First Class Honours | Exchange: Fudan University, Computer Science (2022–23)

Publication

Xiaoxiao YU, L. Du, C. Zeng, L. Deng, H. Luo, X. Gao, J. Tang. *SocialAttributionQA: A Benchmark for Attribution-Evidence-Aware Social Web Question Answering*. **WISE 2026** (CCF B), accepted; Springer LNCS. First author: built a 1,200-question benchmark over 725k Mastodon posts and the ATLAS graph-plus-retrieval solver (**95.3%** accuracy), with ablations and error analysis.

Selected Hackathons & Honors

33 hackathon and competition awards (12 first place or champion); AI and full-stack highlights:

Ava Box — Voice-First Trading Agent

AVE CLAW Hackathon 2026 — Champion

Sole developer | ESP32-S3, C++, Python, LVGL

- Built a handheld voice agent that routes speech to trading tools based on the current screen; every order needs a button confirmation, and the LVGL UI is shared between firmware and a desktop simulator. Extracted into DeviceKit, an open framework where the chain, speech and language models can be swapped, with over-the-air updates.

OutcomeX — AI Delivery & Settlement

HashKey Chain Horizon 2026 — 2nd, DeFi Track

Sole developer | FastAPI, AgentSkillOS, Solidity

- Built a marketplace where buyers pay for an AI deliverable: an agent runtime (AgentSkillOS) does the work with live progress, and payment stays in escrow until the buyer accepts, with refunds and revenue sharing on-chain.

|      |                                                  |      |                                                        |
|------|--------------------------------------------------|------|--------------------------------------------------------|
| 2025 | Zama Bounty, Universal FHEVM SDK — <b>Winner</b> | 2024 | BeWater AI Crypto — <b>1st, Security &amp; Privacy</b> |
| 2025 | ICBC Cup FinTech, Guangdong — <b>First Prize</b> | 2024 | ETH Beijing — <b>1st, Scroll Track</b>                 |
| 2024 | ETH Shanghai — <b>Overall Champion</b>           | 2021 | PaddlePaddle Anti-Fraud ML — <b>1st (solo)</b>         |

Founded LambdaSoc (Edinburgh) and Lambda Web3 (**1,000+ members**); **7,000+** Xiaohongshu followers for engineering content. Languages: Chinese (native), English (professional).